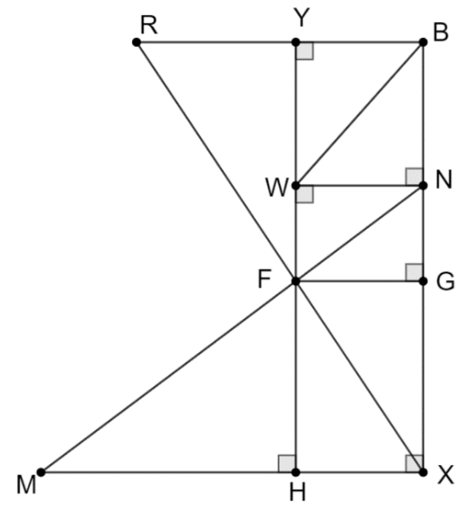
Geometry
Unit 7
Lesson 9 Homework

Name Answer Key_________
Date____________________
Period___________________

A pastry chef is creating a sculpture using chocolate. To ensure the sculpture has a strong base the chef uses a computer program to outline the structure he will use. The outline of the structure is shown.



The table shows known measurements of the structure.

$m\angle FMH = 37°$	$FH = 4$ feet	$GN = 2$ feet
$m\angle XFH = 34°$	$BN = 3$ feet	$m\angle YRF = 57°$

Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure.

	Side	Work	Side Measure
1.	$\overline{HX}$	$tan\, 34 = \frac{HX}{4}$	2.7
2.	$\overline{FM}$	$sin\, 37 = \frac{4}{FX}$	6.6
3.	$\overline{RB}$	$tan\, 57 = \frac{3+2+4}{RB}$	5.8

4. The polygon $FMXBR$ is the base of the chocolate structure. What is the area of the base?

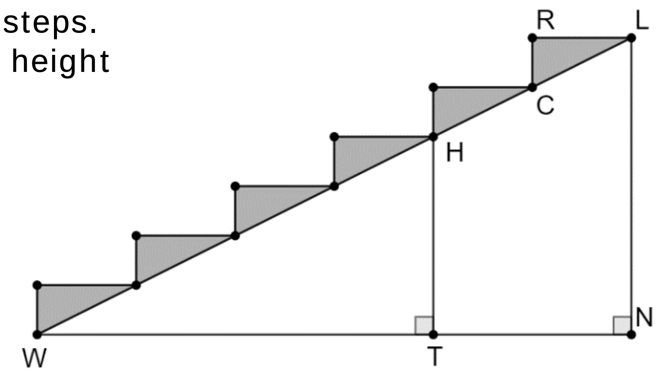
$MH: tan\ 37 = \frac{4}{MH}$
$MH = 5.3$

$\triangle RBX: A = \frac{1}{2} \cdot 9 \cdot 5.8 = 26.1$
$\triangle MFH:\ A = \frac{1}{2} \cdot 5.3 \cdot 4 = 10.6$
$\triangle XFH:\ A = \frac{1}{2} \cdot 2.7 \cdot 4 = 5.4$

$26.1 + 10.6 + 5.4 = 42.1$

Rounding of intermediate steps may result in slight answer variation.

The diagram shows a staircase that consists of 6 steps. The length of each step is 11 inches (in.) and the height each step is 5.25 in.



5. Determine the measure of $\angle CLR$.
Round to the nearest hundredth.
$\tan x = \frac{5.25}{11};\ \tan^{-1}\frac{5.25}{11} = 25.51°$

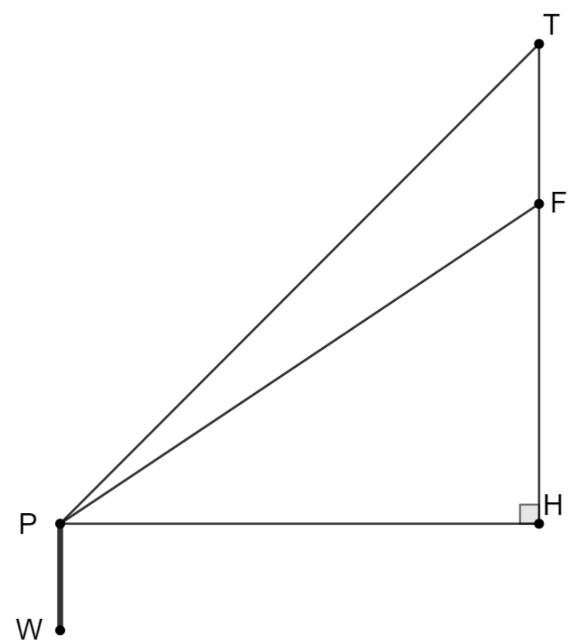
6. Determine the length of $\overline{LW}$.
Round to the nearest hundredth.
$LW = 6 \cdot \sqrt{5.25^2 + 11^2} = 73.13$

7. Determine the measure of $\angle HWT$. Round to the nearest hundredth.
$\angle HWT \cong \angle CLR$ (alternate interior angles)
$m\angle HWT = 25.51°$

8. The homeowner plans to paint quadrilateral $HLNT$ using chalkboard paint. What is the area of quadrilateral $HLNT$? Round to the nearest hundredth.

$$A = \frac{1}{2} \cdot 22 \cdot (31.5 + 21) = 577.50$$

The crown of an oak tree is the area that consists of the branches and leaves of the tree. Leigh Ann was trying to determine the height of an oak tree's crown, TF. The height to Leigh Ann's eye, PW, is 5.5 feet (ft). Leigh Ann is standing 9 feet away from the tree. The angle of elevation from Leigh Ann's eye to the bottom of the oak tree's crown is $34°$. The angle of elevation from Leigh Ann's eye to the top of the oak tree's crown is $45°$.



9. Determine the height, in feet, of the tree. Round to the nearest tenth.

$TH: \tan 45 = \frac{x}{9}; x = 9$
$9 + 5.5 = 14.5$ ft

10. Determine the height, in feet, of the oak tree's crown. Round to the nearest tenth.
$FH: \tan 34 = \frac{x}{9}; x = 6.07$
$14.5 - 6.07 = 8.43$ ft